

Geochemical Characterization of Schirmacher Orthogneiss, Schirmacher Oasis, CDML, East Antarctica

Amit Dharwadkar, M. J. D'Souza, and D. Jayapaul

Antarctica Division, Geological Survey of India, Faridabad

Abstract

The Schirmacher Oasis forms the northernmost outcrop of the polydeformed-polymetamorphosed central Dronning Maud Land, East Antarctica. The most dominant lithounit is quartzo-feldspathic gneiss, which is showing metamorphism up to upper amphibolite facies, has apparently undergone at least 4 phases of deformation. Geochemical characterization of the gneisses reveals their metaluminous to peraluminous nature with enrichment in Fe, Ti, Mg and Ca. They are depleted in Si as compared to average granite composition. These orthogneisses are part of the supracrustals and represent intrusives associated with the 1000 ma crust forming orogeny recognized in Central Dronning Maud Land. These gneisses have been extensively reworked during the later 500 ma Pan-African orogeny.

Keywords: Schirmacher oasis, geochemistry, quartzo-feldspathic gneiss, orthogneiss, orogeny, metamorphism and deformation.

Introduction

Schirmacher Oasis in the central Dronning Maud Land (CDML), East Antarctica is a low lying coastal nunatak covering an area of 35sqkm between longitudes 11°20'E and 11°55'E along latitude 70°45'S. Several workers have carried out geological and structural studies in this area (Ravich and Kamenev, 1975; Kaul et al., 1991; Sengupta, 1988; Ravikant and Kundu,). In this publication an attempt is being made to evaluate geological set up of Schirmacher Oasis on the bases of geochemical studies of orthogneiss mainly the quartzo-feldspathic gneiss and charnockite. Attempt is also made to integrate these results with our understanding of the geological and tectonic evolution of this region. This study is based on field studies carried out during our wintering in Antarctica in year 2001-2002.

Geological Setup

The Schirmacher Oasis marks the northernmost outcrop of CDML. The geological setup of CDML is complex as it has suffered metamorphism and deformation during two prominent tectono-thermal events. Ongoing studies in CDML has indicated that the crust was formed initially during 1000 Ma Grenvillian event which has been extensively reworked during the subsequent Pan African orogeny that caused amalgamation of east and west Gondwanaland.

The Schirmacher Oasis is an important piece of land which may have occupied a key place in the amalgamating land masses. It exposes polydeformed and metamorphosed para and orthogneiss. The dominant grade of metamorphism exhibited by the lithounits is upper amphibolite to granulite facies with widespread indication of retrogression. The peak Ultra High temperature metamorphism has been reported from Schirmacher Oasis. The Orthogneiss is the most dominant lithounit exposed. The orthogneissic rock is dominantly quartzo-feldspathic granitic gneiss, augen gneiss, migmatite, charnockite and enderbites. The charnockite in patches is seen to be retrograding to quartzofeldspathic gneiss. These patches are formed on either side of thin joint planes. Besides the orthogneisses of granitic affinity Schirmacher Oasis also expose in the supracrustals sequence igneous ultramafic and mafic intrusives like granulite, metagabbro, metaultramafites, metanorite etc. his study has been restricted to studying the orthogneiss of granitic affinity.

The rocks of the Schirmacher Oasis have been subjected to at least 4 major deformational episodes accompanied by associated metamorphism. This metamorphism is overprinted on the dominant foliation plane identified as S_1 plane the pervasive regional foliation (ENE-WSW) which was developed during the D2 deformation. A foliation plane earlier to this is indicated by the enclaves. However the D2 deformation event and the associated metamorphism has as such obliterated much of earlier imprint of deformation or metamorphism.

Methodology

Samples of various orthogneisses exposed in the Schirmacher Oasis were collected from different areas (Fig. 1). Total 48 samples were collected during the course of field work. Geochemical analysis for major and few trace elements was carried out at the PPOD Lab., AMSE, GSI, Bangalore, using the XRF technique. The Rare Earths were analysed at the NAA Lab., GSI, Pune. The results of major element analyses are

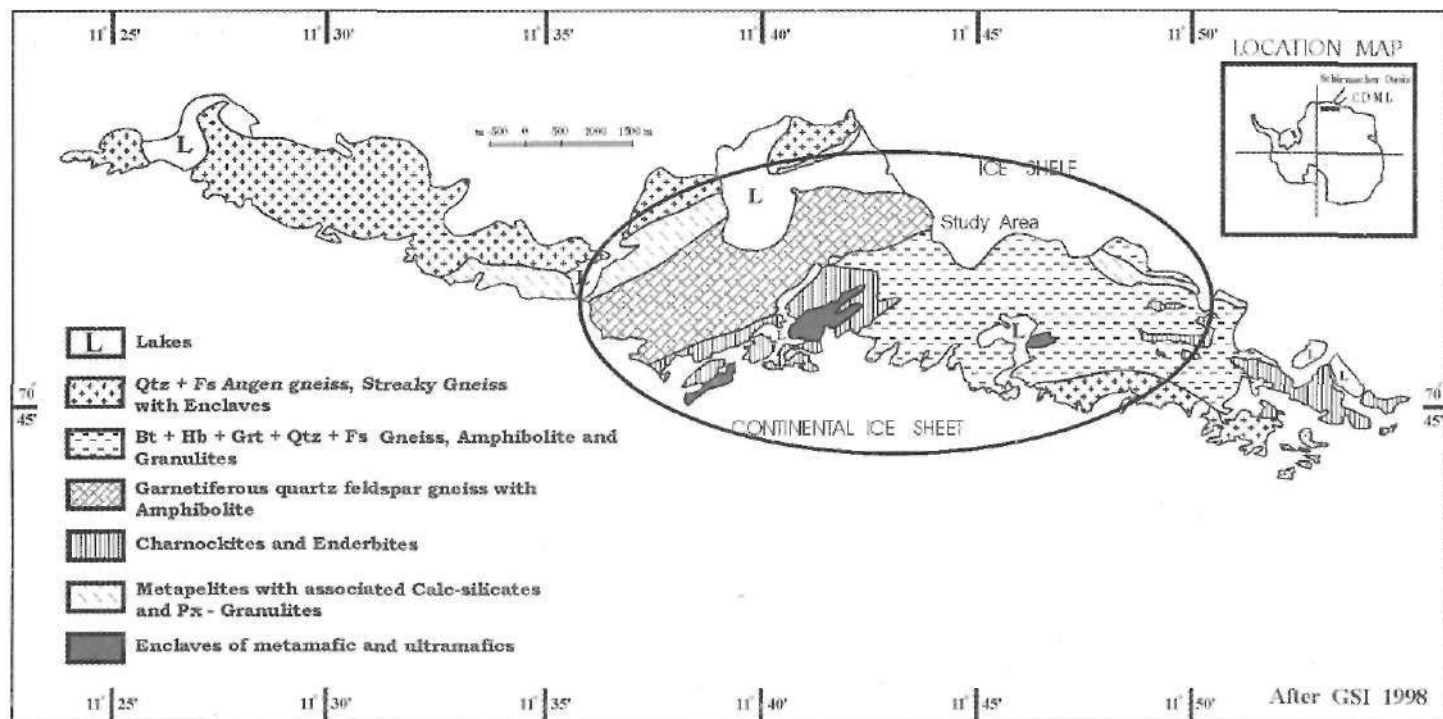


Fig. 1: Geological map of Schirmacher showing the aerial extent of sampled area.

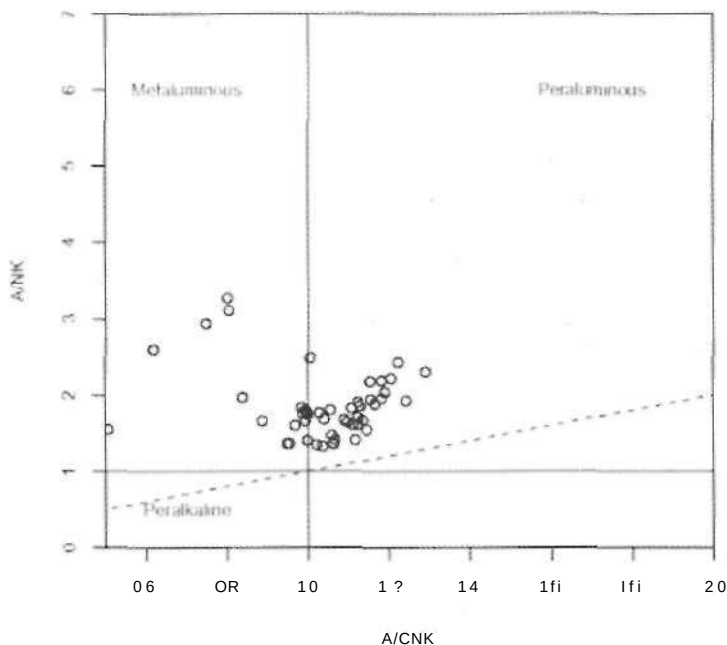


Fig. 2: Most of the gneisses plot in the peraluminous field but some plot in the metaluminous field (after Shand, 1985)

presented in Table 1a. Table 1b presents the REE data while Table 1c gives the CIPW normative data.

Discussion

The average composition of the orthogneisses from Schirmacher Oasis plots in the granite-syenogranite field. The gneisses, in general show depletion in SiO_2 , and alkalis and enrichment in FeO_T , TiO_2 , MgO and CaO as compared to average granite composition. The gneisses are mostly peraluminous to metaluminous (Fig. 2) resulting in the presence of corundum (average 1.4) in the CIPW norms (Table 1c). In the A-F-M diagram (Fig. 3) the Schirmacher gneisses show a calc-alkaline evolutionary trend and appear to be cogenetic in nature. The Schirmacher gneisses show an even spread in the high-K series with a few showing affinity towards shoshonitic series (Fig. 4). Simple correlation plots of Harker type (Fig. 5) reveal that Silica has strong to moderate negative correlation with TiO_2 , Al_2O_3 , FeO , CaO and MgO while FeO has a positive correlation with Al_2O_3 , TiO_2 , CaO and MgO . A negative correlation of FeO with K_2O is observed. There is no mutual correlation between the alkalis (Na_2O and K_2O) with respect to each other.

Table 1: Table showing major oxide, trace element and REE chemical analysis of Schirmacher orthogneiss

Fld_No.	Rock_type	SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	Mn ₂ O	CaO	MgO	Na ₂ O	K ₂ O	P ₂ O ₅	LOI	S	Cr ₂ O ₃	NiO	BaO	Total
XX/GC/1	orthogneiss	70.94	0.55	13.92	4.34	0.05	1.74	1.06	2.98	3.78	0.09	0.18	.	0.05	0.02	0.06	99.76
XX/GC/2	orthogneiss	65.47	0.98	14.45	7.79	0.08	2.82	1.73	2.54	3.46	0.16	0.25	.	0.03	0.01	0.08	99.85
XX/GC/3	leucocratic orthogneiss	70.85	0.47	14.04	4.37	0.06	2.16	0.09	3.25	3.11	0.13	0.27	.	0.06	0.02	0.04	98.92
XX/GC/4	garnetiferous orthogneiss	65.46	0.95	15.1	7.71	0.09	2.9	1.49	2.6	2.9	0.32	.	.	0.09	0.03	0.05	99.69
XX/GC/5	garnetiferous orthogneiss	66.22	1.03	14.81	6.34	0.07	2.62	1.1	2.33	4.62	0.53	0.1	.	0.08	0.03	0.06	99.94
XX/GC/6	garnetiferous orthogneiss	67.37	0.82	14.7	6.39	0.03	2.64	1.46	2.48	3.45	0.22	0.13	.	0.05	0.02	0.07	99.83
XX/GC/7	garnetiferous orthogneiss	67.64	0.86	14.66	6.78	0.07	2.3	1.55	2.56	3.15	0.13	0.1	.	0.05	0.02	0.05	99.92
XX/GC/8	coarse charnockite	62.33	1.28	15.76	7.92	0.78	0.13	2.19	3.35	0.77	0.29	0.11	.	0.05	0.2	.	95.16
XX/GC/9a	orthogneiss	66.56	0.77	14.68	6.18	0.07	2.97	1.3	3.45	2.81	0.23	0.74	.	0.04	0.02	0.05	99.87
XX/GC/9b	orthogneiss	73.45	0.27	13.67	2.18	0.03	1.54	0.5	2.75	5.12	0.06	0.18	.	0.04	0.02	0.08	99.89
XX/GC/9c	orthogneiss	63.13	1.04	15.04	8.33	0.1	3.68	1.68	3.39	2.76	0.32	0.14	.	0.04	0.02	0.06	99.73
XX/GC/9c	charnockite	63.31	0.99	15.16	8.49	0.1	3.94	1.63	3.44	2.39	0.32	.	.	0.05	0.02	0.05	99.89
XX/GC/9f	charnockite	63.78	0.98	15.2	7.98	0.1	3.66	1.61	3.42	2.76	0.31	.	.	0.04	0.01	0.06	99.91
XX/GC/9g	charnockite	63.39	0.95	15.8	7.89	0.09	3.42	1.57	3.38	2.95	0.36	.	.	0.04	0.01	0.08	99.93
XX/GC/10	charnockite	66.73	1.2	15.02	3.67	0.03	1.75	1.43	2.68	6.38	0.25	0.46	.	0.05	0.02	0.11	99.78
XX/GC/11	orthogneiss	71.44	0.61	11.86	5.57	0.05	1.97	2.41	2.84	2.37	0.05	0.19	.	0.03	0.01	0.4	99.8
XX/GC/12	charnockite (enclave)	50.83	0.61	10.75	9.56	0.15	5.45	19.16	1.43	1.02	0.21	0.29	.	0.33	0.05	0.04	99.88
XX/GC/13	orthogneiss (charnockitic)	65.72	1.02	14.29	8.25	0.08	2.63	2.3	2.64	2.72	0.09	.	.	0.04	0.01	0.04	99.84
XX/GC/15	orthogneiss	65.55	0.77	14.54	6.07	0.07	4.19	1.72	3.3	3.08	0.26	0.28	.	.	0.03	0.01	99.87
XX/GC/16	leucocratic orthogneiss	74.22	0.22	12.96	2.57	0.04	1.89	0.44	2.64	4.47	0.06	0.24	.	0.05	0.02	0.07	99.89
XX/GC/17	garnetiferous orthogneiss	63.33	1.28	15.35	8.91	0.09	3.44	1.68	2.76	2.32	0.31	0.15	.	0.05	0.02	0.05	99.74
XX/GC/18	garnetiferous orthogneiss	68.69	0.69	14.15	5.91	0.07	2.37	1.01	2.53	3.81	0.24	0.24	.	0.05	0.02	0.05	99.83

(Contd.)

Table 1: Table showing major oxide, trace element and REE chemical analysis of Schirmacher orthogneiss (Contd.)

Fld. No.	Rock_type	SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	Mn _o	CaO	MgO	Na ₂ O	K ₂ O	P ₂ O ₅	LOI	S	Cr ₂ O ₃	NiO	BaO	Total
XX/GC/19	garnetiferous orthogneiss	71.54	0.51	13.74	4.69	0.05	2.02	0.02	2.59	3.96	0.24	0.36	.	0.02	0.01	0.04	99.79
XX/GC/20	garnetiferous orthogneiss	67.47	0.78	14.32	6.42	0.05	2.21	1.44	2.6	3.99	0.21	0.24	.	0.05	0.02	0.04	99.84
XX/GC/21	garnetiferous orthogneiss (leucocratic)	70.89	0.37	14.02	3.97	0.04	1.43	0.08	2.62	5.22	0.08	0.32	.	0.05	0.02	0.05	99.16
XX/GC/ 21a	garnetiferous orthogneiss (leucocratic)	71.38	0.46	13.09	4.4	0.05	2.94	1.05	2.69	3.47	0.11	0.02	.	0.08	0.03	0.03	99.8
XX/GC/22	garnetiferous orthogneiss	64.94	1.06	15.48	8.09	0.09	3.22	1.54	2.62	2.47	0.32	0.01	.	0.06	0.02	0.05	99.97
XX/GC/23	orthogneiss	72.38	0.04	12.55	3.95	0.05	1.83	1.86	2.62	3.9	0.1	0.08	.	0.06	0.02	0.06	99.51
XX/GC/24	amphibolite	51.97	2.2	15.29	11.81	0.16	8.14	5.14	2.99	1.41	0.57	0.05	.	0.06	0.02	0.02	99.83
XX/GC/25	garnetiferous orthogneiss (charnockitic)	62.55	1.28	15.78	9.66	0.1	3.52	1.95	2.59	2.07	0.28	0.01	.	0.06	0.02	0.07	99.94
XX/GC/ 25a	garnetiferous orthogneiss (biotite rim)	63.49	1.2	15.29	9.07	0.09	3.27	1.91	2.71	2.34	0.27	0.17	.	0.03	0.01	0.08	99.96
XX/GC/ 26	augen gneiss/mylonite	61.51	1.12	15.25	9.69	0.11	3.38	1.22	3.06	3.85	0.48	0.12	.	0.03	0.01	0.08	99.92
XX/GC/27	garnetiferous orthogneiss	65.81	0.93	15.01	7.45	0.06	2.9	1.61	2.79	2.9	0.31	0.01	.	0.11	0.04	0.05	99.98
XX/GC/28	orthogneiss	72.29	0.2	14.94	1.67	0.03	1.93	0.64	4.24	3.83	0.06	0.02	.	0.03	0.01	0.05	99.95
XX/GC/29	charnockite	64.45	0.97	15.07	7.34	0.09	3.73	1.7	3.46	2.48	0.32	0.03	.	0.05	0.02	0.08	99.81
XX/GC/ 29a	orthogneiss	72.89	0.41	13.27	3.56	0.05	1.68	0.44	2.91	4.25	0.13	0.26	0.01	0.04	0.02	0.06	99.98
XX/GC/30	charnockite	49.47	1.56	16.73	11.92	0.16	8.67	7.43	2.56	0.83	0.24	0.26	0.01	0.04	0.02	0.01	99.91
XX/GC/31	augen gneiss/mylonite	64.93	0.84	15.16	6.78	0.08	3.39	1.72	3.41	2.27	0.28	0.26	0.01	0.03	0.02	0.07	99.25
XX/GC/ 31a	granite/diorite	65.87	1.3	14.3	5.67	0.05	2.27	1.15	2.69	5.33	0.56	0.5	.	0.06	0.02	0.08	99.85

(Contd.)

Table 1: Table showing major oxide, trace element and REE chemical analysis of Schirmacher orthogneiss (Contd.)

Fld_No.	Rock_type	SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	Mn _o	CaO	MgO	Na ₂ O	K ₂ O	P ₂ O ₅	LOI	S	Cr ₂ O ₃	NiO	BaO	Total
XX/GC/32	orthogneiss	61.69	1.1	15.52	8.56	0.1	3.65	1.62	3.35	3.12	0.4	0.56	0.01	0.04	0.01	0.07	99.8
XX/GC/33	leucocratic granite	77.29	0.02	12.8	0.62	0.01	1.64	0.1	2.77	4.36	0.05	0.19	.	0.06	0.02	0.05	99.98
XX/GC/34	diorite/dolerite	64.66	1.63	15.08	6.25	0.06	3.75	1.45	3.12	2.69	0.49	0.47	0.02	0.05	0.02	0.11	99.85
XX/GC/36	foliated charnockite	55.26	2.44	14.81	12.78	0.14	5.57	2.63	3.21	2.08	0.81	0.07	0.02	0.06	0.02	0.04	99.94
XX/GC/37	garnetiferous orthogneiss	69.23	0.54	14.43	5.32	0.06	2.44	1.07	2.7	3.72	0.14	0.18	.	0.06	0.02	0.06	99.97
XX/GC/38	garnetiferous orthogneiss	69.89	0.72	14.83	6.46	0.07	2.83	1.16	2.46	3.65	0.32	0.23	0.07	0.08	0.03	0.08	102.9
XX/GC/38a	garnetiferous orthogneiss	66.51	0.82	15.01	6.78	0.08	3.01	1.19	2.59	3.35	0.37	0.4	0.05	0.07	0.02	0.07	100.3
XX/GC/40	orthogneiss?	70.56	0.62	13.73	4.96	0.05	2.02	0.98	2.48	4.24	0.13	0.03	.	0.04	0.02	0.05	99.91
XX/GC/42	orthogneiss	65.88	0.64	14.45	6.06	0.08	2.59	2.08	3.29	4.12	0.3	0.2	.	0.04	0.01	0.1	99.84
XX/GC/44	orthogneiss	65.85	0.95	13.78	7.52	0.1	2.42	0.92	2.89	4.96	0.34	0.04	.	0.05	0.02	0.04	99.88
XX/GC/47/1	orthogneiss	65.68	0.9	14.08	7.14	0.09	2.44	0.9	2.9	5.13	0.35	0.07	.	0.04	0.02	0.05	99.79
XX/GC/47b	anorthosite?	65.68	1.26	15.07	3.9	0.04	2.51	1.17	2.81	6.3	0.43	0.63	.	0.05	0.02	0.09	99.96
XX/GC/47c	orthogneiss (mafic rich)	51.18	1.59	15.23	10.02	0.15	10.11	6.69	3.03	1.15	0.3	0.26	.	0.09	0.02	0.01	99.83
XX/GC/47c1	orthogneiss	50.66	1.48	15.23	10.19	0.15	10.47	6.92	3.03	0.88	0.27	0.26	.	0.11	0.01	.	99.66
XX/GC/48a	charnockite	50.4	1.67	16.2	11.03	0.15	8.89	7.17	2.68	1.02	0.22	0.29	.	0.08	0.02	.	99.82

Table 1B: REE analysis of Schirmacher gneisses

Fld. No	La	Ce	Nd	Sm	Eu	Tb	Yb	Lu	Th	Sc	Co	Cr	Hf	Ta
XX/GC/1	32.00	66.00	30.00	6.00	1.50	1.00	3.30	0.50	9.40	14.00	12.00	551.00	8.00	0.29
XX/GC/2	24.00	42.00	19.00	4.70	2.10	1.00	4.00	0.60	2.20	20.00	16.00	327.00	9.80	0.44
XX/GC/3	25.00	45.00	23.00	5.20	1.30	1.10	4.60	0.70	4.10	14.00	9.00	531.00	5.80	0.14
XX/GC/4	42.00	92.00	52.00	12.00	1.50	2.30	11.00	1.80	1.10	16.00	14.00	665.00	20.00	0.11
XX/GC/5	48.00	95.00	52.00	9.50	1.50	1.50	8.00	1.40	6.60	23.00	12.00	666.00	10.00	0.16
XX/GC/6	34.00	68.00	30.00	7.00	1.70	1.30	6.00	1.00	5.20	19.00	12.00	444.00	10.00	0.43
XX/GC/7	37.00	70.00	32.00	7.00	1.70	1.50	6.50	1.10	8.50	22.00	14.00	475.00	11.00	0.31
XX/GC/8	30.00	66.00	28.00	7.00	1.60	1.20	4.30	0.80	1.60	19.00	21.00	483.00	11.00	0.32
XX/GC/9a	33.00	68.00	30.00	6.00	1.70	0.85	2.60	0.36	1.00	17.00	12.00	367.00	13.00	0.36
XX/GC/9b	57.00	100.00	48.00	6.70	1.70	0.90	2.90	0.39	26.00	7.20	6.50	435.00	4.80	0.22
XX/GC/9c	28.00	65.00	28.00	6.50	2.10	1.10	3.50	0.50	0.27	23.00	15.00	395.00	15.00	0.33
XX/GC/9e	27.00	64.00	27.00	6.40	2.00	1.10	4.40	0.66	0.11	23.00	14.00	405.00	13.00	0.40
XX/GC/9f	30.00	66.00	38.00	9.20	2.40	1.50	4.20	0.60	0.50	25.00	17.00	350.00	18.00	0.10
XX/GC/9g	31.00	70.00	50.00	10.00	2.80	1.70	7.20	1.00	0.40	28.00	17.00	420.00	23.00	0.40
XX/GC/10	200.00	320.00	106.00	23.00	2.30	1.90	2.00	0.25	100.00	37.00	11.00	420.00	35.00	0.20
XX/GC/11	51.00	100.00	50.00	8.30	1.80	1.50	9.00	1.70	13.00	27.00	6.00	300.00	13.00	0.20
XX/GC/12	12.00	24.00	12.00	3.00	1.10	0.70	2.90	0.45	0.88	35.00	60.00	1000+	4.90	0.20
XX/GC/13	30.00	63.00	27.00	5.80	1.60	1.20	7.30	1.20	2.50	28.00	13.00	337.00	12.00	0.15
XX/GC/15	23.00	48.00	27.00	7.00	2.30	1.30	4.80	0.75	1.60	19.00	13.00	310.00	13.00	0.40
XX/GC/16	41.00	80.00	30.00	5.40	1.70	0.90	3.20	0.47	7.90	8.20	5.50	490.00	8.70	0.20

(Contd.)

Table 1B: REE analysis of Schirmacher gneisses (Contd.)

File No	La	Ce	Nd	Sm	Eu	Tb	Yb	Lu	Th	Sc	Co	Cr	Hf	Ta
XX/GC/17	54.00	110.00	56.00	12.00	2.20	1.90	11.00	2.00	3.20	36.00	15.00	470.00	20.00	0.30
XX/GC/18	38.00	80.00	50.00	9.50	1.40	1.30	7.10	1.20	7.60	19.00	12.00	400.00	11.00	0.20
XX/GC/19	29.00	65.00	30.00	6.90	1.50	1.20	7.00	1.20	6.40	16.00	7.30	170.00	9.00	0.10
XX/GC/20	36.00	75.00	47.00	8.50	1.30	1.50	6.50	1.00	9.40	17.00	11.00	500.00	8.50	0.10
XX/GC/21	23.00	40.00	18.00	3.90	1.30	0.94	6.60	1.10	1.10	15.00	8.00	480.00	5.20	0.16
XX/GC/21a	32.00	70.00	30.00	6.10	1.20	1.20	5.20	0.85	6.90	15.00	11.00	738.00	7.40	0.10
XX/GC/22	39.00	84.00	51.00	9.40	1.40	1.80	11.00	1.80	6.80	28.00	13.00	470.00	11.00	0.40
XX/GC/23	26.00	56.00	21.00	4.10	1.30	0.75	3.10	0.45	5.70	13.00	6.20	315.00	7.00	0.27
XX/GC/24	32.00	75.00	48.00	9.30	2.90	1.60	5.70	0.85	1.60	35.00	42.00	482.00	6.40	0.31
XX/GC/25	56.00	119.00	59.00	11.00	2.60	1.80	11.00	2.00	12.00	34.00	16.00	525.00	13.00	0.39
XX/GC/25a	57.00	119.00	62.00	12.00	2.70	1.90	11.00	1.90	14.00	35.00	17.00	345.00	15.00	0.40
XX/GC/26	30.00	73.00	48.00	11.00	3.40	1.90	6.70	1.00	1.10	24.00	13.00	293.00	18.00	0.86
XX/GC/27	40.00	84.00	48.00	6.70	1.90	1.20	6.10	1.00	5.90	22.00	18.00	803.00	9.50	0.42
XX/GC/28	14.00	23.00	9.20	2.00	0.64	0.39	1.60	0.24	5.50	4.30	5.50	318.00	2.70	0.20
XX/GC/29	22.00	46.00	23.00	5.60	2.10	1.20	3.80	0.59	1.40	19.00	15.00	335.00	13.00	0.28
XX/GC/29a	30.00	70.00	33.00	6.20	1.20	1.40	7.40	1.30	3.70	19.00	4.80	357.00	6.50	0.31
XX/GC/30	9.00	21.00	13.00	3.75	1.40	0.84	3.50	0.53	1.20	29.00	56.00	305.00	3.60	0.15
XX/GC/31	21.00	48.00	23.00	5.50	1.80	1.00	3.50	0.54	0.30	17.00	10.00	270.00	10.00	0.62
XX/GC/31a	81.00	160.00	75.00	15.00	2.30	1.00	7.90	1.30	0.90	10.00	11.00	410.00	31.00	2.00
XX/GC/32	36.00	70.00	84.00	9.70	2.30	1.20	4.20	0.80	0.50	25.00	15.00	290.00	20.00	1.20

(Contd.)

Table 1B: REE analysis of Schirmacher gneisses (*Contd.*)

Flt_No	La	Ce	Nd	Sm	Eu	Tb	Yb	Lu	Th	Sc	Co	Cr	Hf	Ta
XX/GC/33	17.00	32.00	15.00	3.90	1.20	0.60	0.92	0.12	0.50	2.10	4.80	490.00	2.20	0.25
XX/GC/34	42.00	96.00	52.00	9.70	2.90	1.20	3.20	0.45	1.00	21.00	18.00	430.00	24.00	1.00
XX/GC/36	33.00	77.00	52.00	15.00	2.90	2.30	7.50	1.00	0.40	32.00	24.00	350.00	15.00	2.50
XX/GC/37	59.00	104.00	53.00	9.70	2.10	1.50	5.80	1.00	14.00	17.00	10.00	390.00	16.00	1.10
XX/GC/38	49.00	67.00	43.00	11.00	2.60	1.90	9.10	1.30	6.30	21.00	12.00	470.00	15.00	1.90
XX/GC/38a	56.00	112.00	57.00	12.00	2.90	2.10	9.10	1.30	16.00	22.00	15.00	530.00	14.00	1.50
XX/GC/40	83.00	143.00	68.00	12.00	3.20	1.90	8.80	1.30	23.00	18.00	10.00	350.00	10.00	2.00
XX/GC/42	45.00	97.00	47.00	5.30	1.60	0.82	2.80	0.42	11.00	18.00	7.30	270.00	18.00	0.52
XX/GC/44	48.00	101.00	54.00	13.00	2.20	2.20	6.90	1.00	3.40	17.00	8.00	403.00	10.00	1.10
XX/GC/47/1	44.00	101.00	54.00	13.00	2.20	1.90	7.10	1.00	5.30	17.00	6.50	340.00	16.00	1.20
XX/GC/47b	45.00	97.00	50.00	7.40	2.00	0.86	2.20	0.32	2.50	4.40	11.00	348.00	27.00	0.76
XX/GC/47c	17.00	35.00	19.00	4.60	1.50	0.92	3.60	0.56	1.50	38.00	33.00	606.00	4.60	0.20
XX/GC/47c1	20.00	40.00	22.00	6.10	2.70	1.20	4.10	0.64	2.60	45.00	40.00	830.00	5.40	0.79
XX/GC/48a	17.00	37.00	20.00	4.90	1.70	0.96	3.40	0.50	1.90	33.00	46.00	594.00	4.30	0.68

Table 1C: CIPW Normative data for Schirmacher gneisses

	Q	C	Or	Ab	An	Di	Hf	Il	Hm	Tn	Ru	Ap	Sum
GC1	34.08	1.98	22.34	25.22	8.04		2.64	0.11	4.34		0.49	0.21	99.46
GC2	29.28	1.78	20.45	21.49	12.94		4.31	0.17	7.79		0.89	0.38	99.49
GC3	34.44	1.71	18.38	27.5	9.87		2.24	0.13	4.37		0.4	0.31	99.35
GC4	31.71	3.18	17.14	22	12.3		3.71	0.19	7.71		0.85	0.76	99.54
GC5	29.23	2.48	27.3	19.72	9.54		2.74	0.15	6.34		0.95	1.26	99.7
GC6	32.53	2.61	20.39	20.99	11.66		3.64	0.06	6.39		0.79	0.52	99.57
GC7	33.82	3.17	18.62	21.66	10.56		3.86	0.15	6.78		0.78	0.31	99.71
GC8	26.46	0.78	4.55	28.35	23.56		5.45	1.67	7.92		0.4	0.69	99.82
GC9a	28.08	1.11	16.61	29.19	13.23		3.24	0.15	6.18		0.69	0.54	99.03
GC9b	33.98	0.95	30.26	23.27	7.25		1.25	0.06	2.18		0.24	0.14	99.57
GC9c	23.36	0.55	16.31	28.69	16.17		4.18	0.21	8.33		0.93	0.76	99.49
GC9e	24.18	0.52	14.12	29.11	17.46		4.06	0.21	8.49		0.88	0.76	99.79
GC9f	23.96	0.67	16.31	28.94	16.13		4.01	0.21	7.98		0.87	0.73	99.82
GC9g	23.79	1.69	17.43	28.6	14.61		3.91	0.19	7.89		0.85	0.85	99.82
GC10	21.55	1.12	37.7	22.68	7.05		3.56	0.06	3.67		1.17	0.59	99.15
GC11	38.18	1.16	14.01	24.03	9.45		6	0.11	5.57		0.55	0.12	99.17
GC12	0.2		6.03	12.1	19.9	3.29	46.2	0.32	9.56	1.08	0	0.5	99.18
GC13	31.07	2.41	16.19	22.34	12.46		5.73	0.17	8.25		0.93	0.21	99.77

(Contd.)

Table 1C: CIPW Normative data for Schrmacher gneisses (Contd.)

	Q	C	Or	Ab	An	Di	Hy	Il	Hm	Tn	Ru	Ap	Sum
GC15	24.47		18.2	27.92	15.76	0.71	3.95	0.15	6.07	1.7	0	0.62	99.55
GC17	29.4	2.79	13.71	23.35	15.04		4.18	0.19	8.91		1.18	0.73	99.49
GC18	33.49	2.13	22.52	21.41	10.19		2.52	0.15	5.91		0.61	0.57	99.48
GC19	37.64	2.09	23.4	21.92	8.45		0.05	0.11	4.69		0.45	0.57	99.37
GC20	30.79	2.21	23.58	22	9.59		3.59	0.11	6.42		0.72	0.5	99.5
GC21	31.64	1.65	30.85	22.17	6.57		1.99	0.09	3.07		0.32	0.19	98.54
GC21a	35		20.51	22.76	13.39		2.62	0.11	4.4	0.33	0.27	0.26	99.65
GC22	21.95	3.41	14.6	22.17	13.88		3.84	0.19	8.09		0.96	0.76	89.85
GC23	35.8	0.93	23.05	22.17	8.43		4.63	0.11	3.95		0.34	0.24	99.65
GC25	29.9	3.55	12.23	21.92	15.63		4.86	0.21	9.66		1.17	0.66	99.8
GC25a	29.68	3	13.83	22.93	14.46		4.76	0.19	9.07		1.1	0.64	99.66
GC26	21.27	1.05	22.75	25.89	13.63		3.04	0.24	9.39		1	1.14	99.4
GC27	29.96	2.08	17.14	23.61	14.18		4.01	0.13	7.45		0.86	0.07	99.49
GC28	28.05	0.45	22.63	35.88	9.18		1.59	0.06	1.67		0.17	0.14	99.83
GC29	25.21	0.68	14.66	29.28	16.41		4.23	0.19	7.34		0.87	0.76	99.63
GC29a	35.81	1.14	25.12	24.62	7.49		1.1	0.11	3.56		0.35	0.31	99.6
GC30	4.53		4.91	21.66	31.71	3.84	16.73	0.34	11.92	3.39		0.57	99.58

(Contd.)

Table 1C: CIPW Normative data for Schirmacher gneisses (Contd.)

	Q	C	Or	Ab	An	Di	Hy	Il	Hm	Tn	Ru	Ap	Sum
GC31	25.53	1.08	16.25	28.85	14.99		4.28	0.17	6.78		0.75	0.66	99.36
GC31a	24.83	1.32	31.5	22.76	7.6		2.86	0.11	5.67		1.24	1.33	99.22
GC32	21.16	0.95	18.44	28.35	15.49		4.04	0.21	8.56		0.99	0.95	99.13
GC36	15.59		12.29	27.16	19.86		6.55	0.3	12.78	1.75	1.57	1.92	99.77
GC37	32.86	1.86	21.98	22.85	11.19		2.67	0.13	5.32		0.47	0.33	99.66
GC38	31.72	2.45	21.57	20.82	11.95		2.89	0.15	6.46		0.64	0.76	99.41
GC38a	31.44	2.54	19.8	21.92	12.52		2.96	0.17	6.78		0.73	0.88	99.73
GC40	34.48	1.7	25.06	20.99	9.17		2.44	0.11	4.96		0.56	0.31	99.78
GC44	24.46	0.07	29.31	24.45	9.78		2.29	0.21	7.52		0.84	0.81	99.75
GC47/1	23.6	0.16	30.32	24.54	9.82		2.24	0.19	7.14		0.8	0.83	99.63
GC47c1	2.93		5.2	25.64	25.87	15.36	10.11	0.32	10.19	3.22		0.64	99.49
GC48a	4.71		6.03	22.68	29.16	6.44	14.87	0.32	11.03	3.68		0.52	99.44

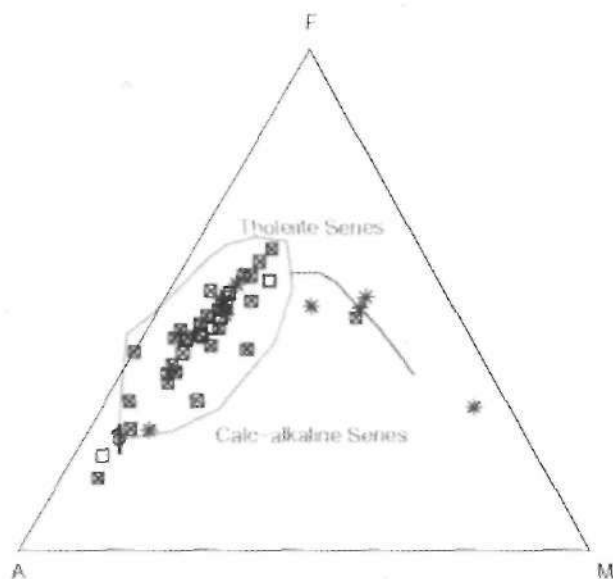


Fig.3: A-F-M plot of Schirmacher gneisses showing calc-alkaline trend of evolution.

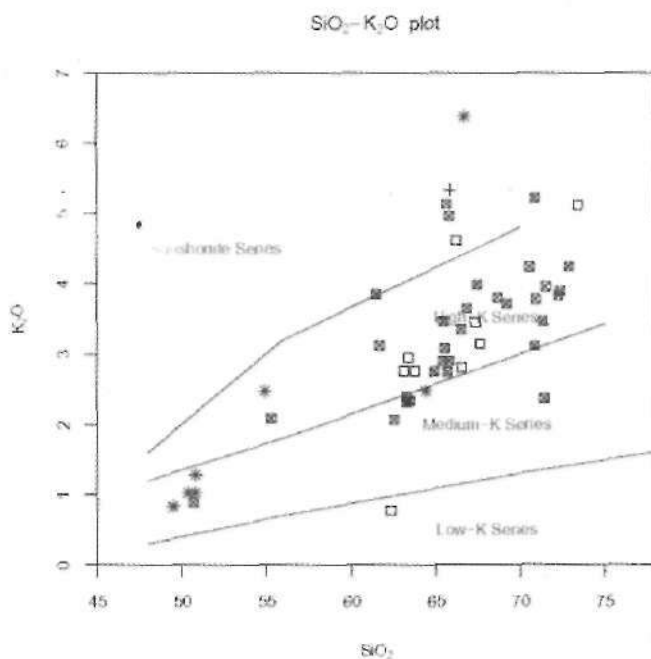


Fig. 4: Most of the gneisses plot in the high-K series with spillover into shoshonitic series. The charnockitic rocks plot in the medium-K series (after Piccerello, 1976).

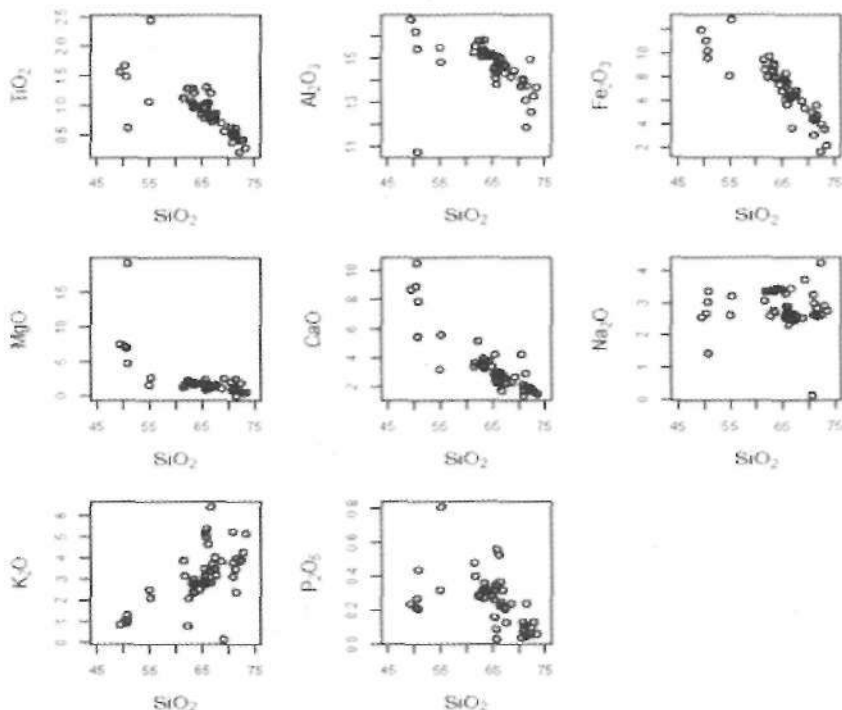


Fig. 5: Harker type variation diagram of various major oxides against silica.

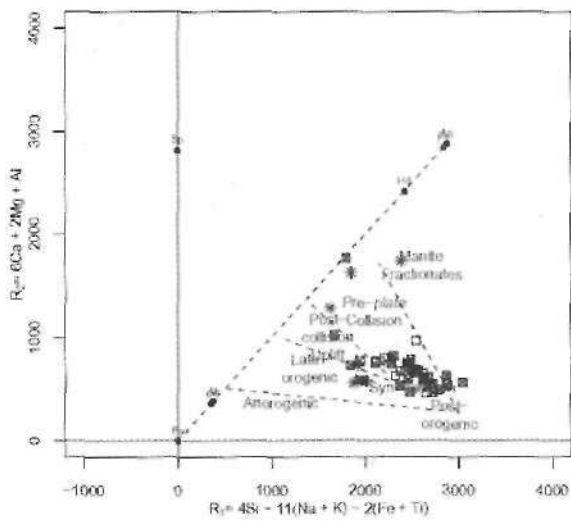


Fig. 6: Multicationic tectonic discrimination plot after Batchelor and Bowden, 1985.

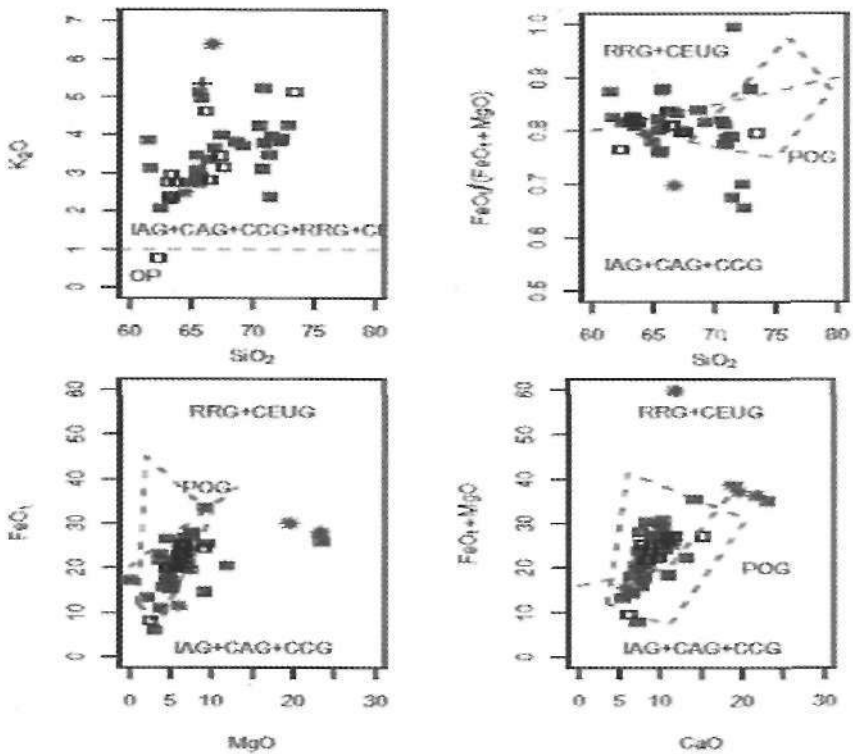


Fig. 7: Major element tectonic discrimination plot after Maniar & Piccoli, 1984, showing post orogenic signature.

The R1-R2 multicationic discriminant plot by Batchelor and Bowden, 1985, suggests a syn to late tectonic origin for the Schirmacher gneisses (Fig. 6). This observation is also supported by independent field studies and metamorphic history of these gneisses. In the Maniar and Piccoli (1984) major element discrimination diagram, the Schirmacher gneisses plot in the RRG+CEUG+CCG field and the post orogenic field indicating that these gneisses have a mixed origin with substantial contribution from lower crust (Fig. 7). Since the Schirmacher orthogneisses have been emplaced over a long temporal as well as spatial regime, it is quite likely that they will exhibit mixed chemical signatures. The REE data shows that these orthogneisses have low La/Lu ratio indicating their less evolved nature. The gneisses show a medium negative Eu anomaly (Fig. 8) and show affinity somewhere between upper and lower crust. The geochemical characteristics of the Schirmacher gneisses indicate that they have been significantly contributed by reworking of the older crust. They have been emplaced in a syn to late tectonic setup at moderate depths.

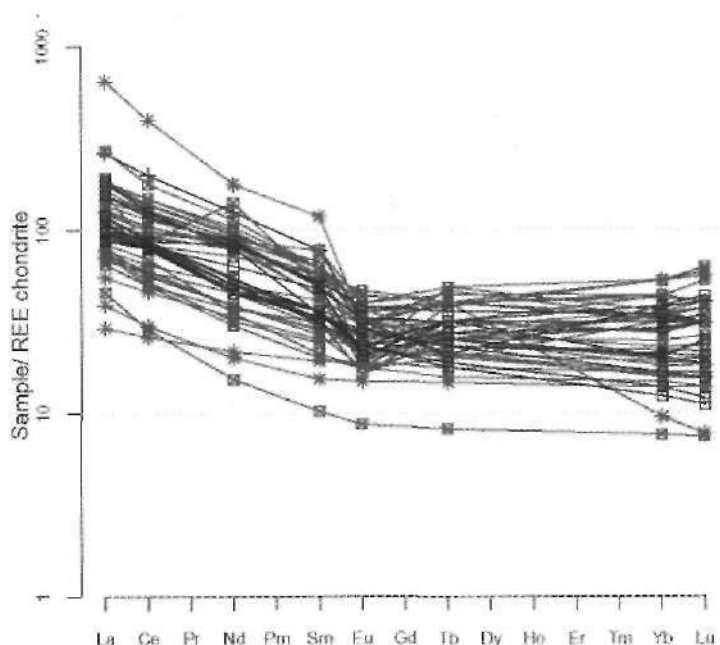


Fig. 8: REE spidergram of Schirmacher gneisses (chondrite normalized).

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